

Phenotypic and genetic associations with colostrum yield and composition in dairy cows

Limited knowledge exists on what affects colostrum yield and composition in dairy cattle. This study investigated phenotypic and genetic factors associated with colostrum yield and composition in cows. Data from 3,681 genotyped Holstein cows from 2 farms (1 = 856; 2 = 2,825) and 4,205 colostrum samples were included in the study. Colostrum yield and concentrations of fat, true protein, total solids, and immunoglobulin (**Ig**) G were evaluated. Information for all cows included parity group prepartum, calving season, body condition score prepartum, gestation length, calf sex and birth body weight (**BW**), dry period length, and days in the prepartum diet. Cows in farm 1 were individually fed prepartum and intake of dry matter (**DM**, kg/d), net energy (**NE**, Mcal/d), metabolizable protein (**MP**, g/d) were evaluated. Data were analyzed using linear mixed-effects models with the MIXED procedure of SAS. Univariable models selected predictors with $P \leq 0.20$. Multivariable models were built and a backward stepwise elimination was performed based on model fit statistics (AICc) and $P > 0.10$. All models included the random effect of treatment nested within farm and for nutrient intake models, the random effects of treatment nested within experiment and the interaction between nutrient intake and treatment nested with experiment was added. Genetic parameters were estimated and genome-wide associations conducted to identify SNPs and genomic regions associated with colostrum traits. Yield of colostrum (kg) was associated ($P < 0.05$) with calving season (spring=7.0, summer=7.2, fall=6.1, winter=6.5), parity prepartum (nulliparous=6.6, primiparous=7.3, multiparous=6.2), gestation length ($\beta = 0.08 \pm 0.01$), calf BW ($\beta = 0.12 \pm 0.02$), calf sex (female=5.9, male=6.6, twins=7.7), and days dry ($\beta = 0.11 \pm 0.009$). Intake of DM ($\beta = -0.11 \pm 0.09$), NE ($\beta = -0.08 \pm 0.06$), and MP ($\beta = -0.001 \pm 0.0009$) were not associated ($P > 0.10$) with colostrum yield, although MP balance ($\beta = -0.001 \pm 0.0006$) tended ($P = 0.07$) to be negatively associated. Content of IgG (g/L) in colostrum was affected ($P < 0.05$) by calving season (spring=93, summer=90, fall=103, winter=97), parity (nulliparous=93, primiparous=84, multiparous=111), and days dry ($\beta = -0.21 \pm 0.10$). Likewise, intake of DM ($\beta = 0.24 \pm 0.55$), NE ($\beta = 0.22 \pm 0.34$), and MP ($\beta = 0.01 \pm 0.006$) were not associated ($P > 0.10$) with IgG content, whereas MP balance ($\beta = 0.01 \pm 0.006$) tended ($P = 0.08$) to be positively associated. Heritability ($\pm$ SE) estimates for colostrum yield, fat content, protein content, IgG content, and total IgG were 0.30(0.03), 0.31(0.03), 0.34(0.03), 0.28(0.03), and 0.23(0.03), respectively.

Key words: Colostrum, cow, immunoglobulin G